The Measurement Protocol Decides the Conclusion: A Methodological Study of CNN–Vision Transformer Adversarial Robustness Comparisons

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Abstract*—Comparisons of adversarial robustness across architecture families report conflicting conclusions, and the disagreement is usually attributed to models, training recipes, or datasets. We show that a substantial part of it comes from the measurement protocol itself. Using adversarially trained ResNet-18 and ViT-Tiny pairs on CIFAR-10—three independent training runs per architecture, full-test-set evaluation with per-sample logging, and a fixed validation split held out before any training—we vary only how robustness and transfer are measured. Across four established conditioning protocols the measured CNN-to-ViT transfer asymmetry ranges from 4.4 to 14.6 points, a 3.3-fold spread, and the choice decides whether an equivalence test is accepted or rejected, although the direction of the effect is stable. Replicating the entire design on CIFAR-100 widens the per-seed spread from 10.5 to 13.6 points and preserves the sign in all twelve measurements, so the effect is not specific to one dataset. A further ablation that holds the training trajectory fixed and varies only the offline checkpoint-selection rule moves reported robust accuracy by 1.6 to 2.9 points, a fourth source of variance that is almost never reported. A 3x3 matrix with a WideResNet-28-10 reference shows the deviation of unconditioned from conditioned rates is almost entirely explained by the target's clean error (r = 0.997 over six directions that sample only three distinct targets): raw transfer rates are not a valid basis for architecture comparison. A conditional decomposition separates the CNN advantage into a clean-accuracy factor and a conditional-sensitivity factor; the attribution between the two changes between training runs even though the absolute ordering does not. Finally, three claims common in this literature fail to replicate under direct measurement: input gradients show no spatial localization difference, CLS attention entropy does not change under attack at n = 1000, and a ViT-specific transfer attack yields no transfer gain in this regime. Scale-invariant gradient sparsity differences and architecture-specific depth profiles of adversarial damage do survive. We close with reporting requirements for cross-architecture robustness studies.

***Index Terms*—Adversarial robustness, convolutional neural networks, evaluation methodology, deep learning security, reproducibility, transfer attacks, Vision Transformer**

I. Introduction

Deep neural networks are vulnerable to adversarial examples, carefully crafted perturbations that cause confident misclassification while remaining bounded by a small norm budget [1], [2]. As Vision Transformers (ViTs) [3], [4] joined convolutional neural networks (CNNs) [5] as a dominant architecture family, a natural question followed: which family is more robust, and why?

The literature answers this question inconsistently. Bai et al. [6] report that under matched training recipes transformers do not surpass CNNs, whereas Benz et al. [7] report ViTs to be more robust for standardly trained models; studies of cross-architecture transfer likewise disagree on whether attacks move more easily from CNNs to ViTs or the reverse [8]–[10]. These disagreements are usually attributed to differences in models, training recipes, or datasets, and the standard response is to compare under a matched recipe.

This paper makes a different observation. Even when the models, the data, the threat model, and the attack budget are all held fixed, the conclusion still depends on a choice that is rarely reported: the *measurement protocol*. Transfer studies must decide which samples enter the denominator—all test samples, those the target classifies correctly, those both models classify correctly, or those on which the source attack succeeded—and robustness comparisons must decide whether to report an absolute score or to separate it into clean accuracy and conditional susceptibility. These choices are usually treated as reporting details. We show that they are part of the result.

Our setting is deliberately narrow so that the measurement question is isolated. We adversarially train ResNet-18 and ViT-Tiny on CIFAR-10 [11] under a matched objective and threat model, with three independent training runs per architecture and a fixed validation split held out before *any* training stage, so that no model is selected on data it has seen. All evaluations run on the full test set with per-sample outcome logging, which makes paired statistics possible: the same samples can be re-weighted under different protocols without re-running an attack. We then vary the protocol, not the models, and measure how far the conclusion moves.

We also treat three claims that recur in architecture-comparison papers as measurable hypotheses rather than as interpretation: that CNN input gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that ViT-specific transfer attacks improve cross-architecture transfer. Each is tested directly with a purpose-built measurement.

Our contributions are as follows.

1) **Protocol sensitivity, quantified.** On identical models and data, the measured cross-architecture transfer asymmetry ranges from $+4.4$ to $+14.6$ points across four established conditioning protocols, a 3.3-

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fold spread, whereas retraining both architectures from scratch moves the same quantity by under 1.5 points. The protocol decides whether an equivalence test is accepted or rejected; the direction of the asymmetry, unlike its magnitude, is stable.

2) **The effect replicates on a second dataset, and a fourth variance source appears.** Repeating the full design on CIFAR-100 widens the mean per-seed protocol spread to 13.58 ± 1.71 points and preserves the sign in all twelve measurements. Separately, holding a training trajectory fixed and varying only the offline checkpoint-selection rule moves reported PGD-10 accuracy by 1.58 to 2.85 points, which places selection alongside seeds, attacks, and protocols as a reportable source of variance.
3) **Unconditioned transfer rates are confounded, and we measure by how much.** In a 3×3 transfer matrix that adds a WideResNet-28-10 reference, the deviation of raw from conditioned rates is explained almost entirely by the target's clean error ($r = 0.997$ over the six off-diagonal directions, which sample only three distinct targets; we report this as a point estimate without a confidence interval). We also find that incoming transfer tracks the target's own vulnerability rather than the source architecture ($r = 0.986$ across three targets), which reframes the asymmetry as a property of the weaker target rather than of stronger CNN-crafted attacks.
4) **A conditional decomposition, and evidence that its attribution is unstable.** Absolute robustness factors exactly into clean accuracy and conditional susceptibility. Reporting both matters, because the attribution between them changed between our earlier single-run checkpoints and three fresh runs, while the absolute ordering did not.
5) **Three negative results.** Scale-invariant sparsity differences do *not* correspond to spatial localization differences (four spatial measures, three of them null); CLS attention entropy does not change measurably under attack at $n = 1000$; and Token Gradient Regularization [12], a ViT-specific transfer attack, gives no transfer gain in this regime, losing to plain MI-FGSM under a matched budget.
6) **What survives.** Gradient sparsity differences reproduce across independent checkpoint sets, are created by adversarial training rather than by architecture, and are accompanied by an alignment difference that exists only in absolute cosine. Adversarial damage accumulates at qualitatively different depths in the two architectures, and the measured drift profile depends on the token aggregation used to compute it.

We close with a short set of reporting requirements that follow from these measurements.

Section II reviews adversarial evaluation practice and architecture-comparison studies. Section III describes the training protocol, the conditioning protocols, and the measurements. Section IV reports results. Section V discusses implications and limitations, and Section VI concludes.

II. Related Work

A. Adversarial Attacks

Adversarial attacks aim to find minimal perturbations that cause misclassification. The Fast Gradient Sign Method (FGSM) [2] generates perturbations in a single step by following the sign of the loss gradient, with the perturbation magnitude bounded under the L_∞ norm. Projected Gradient Descent (PGD) [13] extends FGSM through iterative optimization, taking multiple gradient-sign steps and projecting back onto the ϵ-ball after each step (formal definitions in Section III-C).

The Carlini & Wagner (C&W) attack [14] formulates adversarial example generation as an optimization problem, often achieving higher success rates but at increased computational cost. More recently, AutoAttack [15] combines multiple attack strategies (APGD-CE, APGD-T, FAB-T, Square Attack) to provide a reliable robustness evaluation, becoming the de facto standard for benchmarking.

B. Defense Mechanisms

Adversarial training (AT) [13] remains the most effective defense, training models on adversarial examples generated during each batch:

$$\min_{\theta} \mathbb{E}_{(x,y)\sim\mathcal{D}} \left[\max_{\|\delta\|_\infty \leq \epsilon} \mathcal{L}(f_\theta(x+\delta), y) \right] \tag{1}$$

TRADES [16] provides a theoretically-grounded trade-off between natural accuracy and robustness:

$$\mathcal{L}_{\text{TRADES}} = \mathcal{L}_{CE}(f(x), y) + \beta \cdot \text{KL}(f(x) \| f(x_{adv})) \tag{2}$$

where β controls the trade-off parameter.

MART [17] addresses the issue of misclassified examples by emphasizing them during training. Other approaches include input transformation [18], certified defenses via randomized smoothing [19], and ensemble methods [20]. Recent advances in certified robustness include Adaptive Randomized Smoothing [21], which improves certification radii through multi-step defenses.

C. CNN Robustness

Extensive research has characterized the adversarial robustness of CNN architectures. Madry et al. [13] demonstrated that adversarial training with PGD attacks provides strong empirical robustness. The RobustBench leaderboard [22] tracks state-of-the-art CNN robustness, with WideResNet variants achieving the highest robust accuracy on CIFAR-10.

Studies have shown that model capacity correlates with achievable robustness [13], with wider and deeper networks demonstrating improved robust accuracy. Architecture-specific findings include the role of skip connections in adversarial transferability [23] and the distinct batch-normalization statistics that adversarial training induces at scale [24]; relatedly, auxiliary batch-normalization branches allow adversarial examples to improve clean accuracy [25]. Adversarial training is also known to reshape input-gradient structure itself, yielding more perceptually aligned [26] and sparser [27] gradients.

D. Vision Transformer Robustness

Vision Transformers [3], building on the self-attention mechanism introduced by Vaswani et al. [4], process images as sequences of patches; their internal representations differ measurably from CNNs' layer by layer [28]. Initial studies suggested that ViTs exhibit some inherent robustness properties due to their global receptive field [29]. Bai et al. [6] systematically investigated whether transformers are more robust than CNNs, finding that under fair training recipes transformers are not more adversarially robust than CNNs, while exhibiting superior robustness under distribution shift attributed to self-attention; Benz et al. [7] reported an early direct comparison including MLP-Mixer and linked the differences to frequency preferences. Our work extends this line of inquiry by analyzing adversarially trained models of comparable scale and providing behavioral characterizations through gradient characteristics and feature degradation analysis. Subsequent work revealed that ViTs remain vulnerable to adversarial attacks, particularly gradient-based methods [8], with patch-targeted attacks exploiting the attention mechanism directly [30] and attention shown to shift measurably under patch perturbations [31].

Shao et al. [32] demonstrated that adversarial training can improve ViT robustness—and that ViT features rely less on high-frequency content—though challenges remain due to training instability. ViT robustness to common corruptions and occlusions has also been characterized [33]. More recently, Jain et al. [34] proposed Adaptive Attention Scaling Adversarial Training (AAS-AT) to address ViT-specific training challenges, achieving state-of-the-art robustness on CIFAR-10 and ImageNet-100; with diffusion-augmented adversarial training, ViTs have even been reported to surpass WideResNets on small-scale datasets [35]. Zhu et al. [36] strengthened transferable attacks on ViTs through gradient normalization scaling and high-frequency adaptation. Despite these advances, comprehensive behavioral comparisons between CNN and ViT robustness under unified evaluation frameworks remain limited.

E. Transfer Attacks

Adversarial examples often transfer between models, enabling black-box attacks [37]. The transferability depends on factors including model architecture, training data, and attack methods [38]. Foundational transfer studies already evaluate transferability on correctly classified samples [39], and standard attack papers condition their success rates for the same reason [40]; recent evaluation guidelines make such protocol choices explicit [41]. In ensemble design, Ravikumar et al. [9] restrict transfer measurement to samples correctly classified by both models precisely to account for accuracy differences between them—the same confound our conditioned metric controls.

Cross-architecture transfer between CNNs and ViTs has received increasing attention. Mahmood et al. [8] measured cross-architecture transfer with a both-correct protocol and found notably low CNN↔ViT transferability; Naseer et al. [10] showed that adversarial examples crafted *on* ViTs transfer poorly under conventional attacks and proposed self-ensemble and token refinement to improve ViT-sourced transferability, and Wei et al. [42] improved ViT-sourced transfer by bypassing attention gradients and patch-wise dropout. Building on these, Token Gradient Regularization (TGR) [12] crafts transferable adversarial samples by exploiting ViT's attention components, while Momentum Integrated Gradients (MIG) [43] achieves improved transferability across both ViTs and CNNs. Chen et al. [44] systematically analyze the factors affecting cross-architecture transferability, finding that architectural inductive biases—CNNs preferring high-frequency information while ViTs favor low-frequency patterns—significantly impact transfer success. Understanding these transfer patterns is crucial for developing robust ensemble systems and evaluating real-world attack scenarios.

F. Research Gap

While individual studies have examined CNN and ViT robustness separately, and the pitfalls of unfair CNN–transformer comparisons have been highlighted before [6], [45], existing comparative work typically reports robustness scores in isolation—often with inconsistent perturbation budgets, attack configurations, or model scales—rather than pairing the comparison with behavioral analysis. A recent study in this journal compares ViT and CNN adversarial robustness directly [46]; our work differs by evaluating on the full test set with AutoAttack and paired statistics, by controlling transfer measurement for clean-accuracy differences, and by accompanying the comparison with gradient- and feature-level behavioral analyses.

Our work engages the measurement question directly and makes it the object of study. Conditioning transfer evaluation on correctly classified samples is established practice [8], [9], [39], [40], and individual papers pick one protocol and proceed. What is missing is a controlled estimate of how much that pick matters. We supply it by holding the models, the data, the threat model, and the attack budget fixed and varying only the protocol, and we find that the measured asymmetry moves by 10.45 ± 0.76 points across the four protocols—a 3.3-fold spread—while retraining both architectures moves that same quantity by under 1.5 points. We further quantify the confound in unconditioned rates with a third architecture ($r = 0.997$ against the target's clean error, over three targets), which turns a known qualitative caveat into a number that can be checked against published results.

Three further claims recur in this literature that our design lets us test rather than assume: that CNN gradients are spatially more localized, that adversarial perturbation degrades ViT attention, and that transformer-specific transfer attacks [10], [12] improve cross-architecture transfer. We measure each directly, and none holds in our setting—results we report because an untested assumption of this kind is exactly what a protocol-sensitive comparison cannot afford.

III. Methodology

A. Threat Model

We consider the standard L_∞ threat model where an adversary can perturb input images within a bounded region:

$$\mathcal{B}_\epsilon(x) = \{x' : \|x' - x\|_\infty \leq \epsilon\} \tag{3}$$

Following established conventions for CIFAR-10 [11], [22], we use $\epsilon = 8/255 \approx 0.031$ as the primary perturbation budget, with additional full-test PGD-10 evaluations at $\epsilon \in \{2/255, 4/255, 16/255\}$ for the epsilon sweep analysis.

We evaluate under both white-box (full model access) and black-box (transfer attack) scenarios. For white-box attacks, the adversary has complete knowledge of model architecture and parameters. For transfer attacks, adversarial examples are generated on a source model and evaluated on a different target model.

B. Model Architectures

a) CNN Models: **ResNet-18** [5]: A widely-used baseline CNN with residual connections, containing 11.2M parameters. We use the standard architecture designed for CIFAR-10 (initial 3×3 convolution instead of 7×7).

WideResNet-28-10 [47]: A wider variant of ResNet with 36.5M parameters, representing the state-of-the-art architecture for adversarial robustness on CIFAR-10. We use the pretrained robust checkpoint of Gowal et al. [48] distributed via RobustBench [22]; note that this model was trained with additional unlabeled data and a stronger recipe than our standard adversarial training, so it serves as a reference point rather than a matched comparison.

b) Vision Transformer Models: **ViT-Tiny**: We use the ViT-Tiny architecture from the `timm` library [49], adapted for CIFAR-10 by bilinearly upsampling inputs to 224×224. The architecture uses 16×16 patches (196 patches per image), 192-dimensional embeddings, 12 transformer blocks with 3 attention heads each, MLP ratio of 4, and totals 5.7M parameters. The model is trained *from scratch* on CIFAR-10 (no ImageNet pretraining) with basic augmentation (random crop and horizontal flip), then adversarially fine-tuned. Since ViT robustness on small datasets is known to be sensitive to the training recipe [50], [51], our architecture-level conclusions should be interpreted as holding *under this baseline training recipe* rather than for optimally trained ViTs.

C. Attack Methods

a) FGSM: Single-step attack with step size equal to the perturbation budget:

$$x_{adv} = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}_{CE}(f(x), y)) \tag{4}$$

b) PGD: Iterative attack with step size $\alpha = 2/255$ and random initialization:

$$x^{t+1} = \Pi_{\mathcal{B}_\epsilon(x)}\left(x^t + \alpha \cdot \text{sign}(\nabla_{x^t} \mathcal{L}_{CE}(f(x^t), y))\right) \tag{5}$$

with $x^0 = x + \mathcal{U}(-\epsilon, \epsilon)$. We use PGD-10 (10 iterations) for both adversarial training and evaluation throughout this work (Table I).

c) AutoAttack: We use AutoAttack [15], the standardized ensemble attack combining APGD-CE (Auto-PGD with cross-entropy loss), APGD-T (Auto-PGD with targeted DLR loss), FAB-T (targeted Fast Adaptive Boundary attack), and Square Attack (score-based black-box attack). This ensemble is a standardized strong benchmark for robustness evaluation.

D. Defense Methods

a) Adversarial Training (AT): We employ a mixed adversarial training objective that combines clean and adversarial losses:

$$\mathcal{L} = (1 - \lambda)\,\mathcal{L}_{CE}(f_\theta(x), y) + \lambda\,\mathcal{L}_{CE}(f_\theta(x_{adv}), y) \tag{6}$$

where $\lambda = 0.5$ balances standard classification performance with adversarial robustness. The inner maximization generates adversarial examples using PGD-10 ($\alpha = 2/255$, $\epsilon = 8/255$), computed with the model in training mode so that attack generation and the parameter update see consistent batch-normalization statistics, following standard adversarial-training implementations [52]. This mixed objective, widely adopted in practice, helps maintain clean accuracy while building robustness, as training solely on adversarial examples can degrade natural performance. While alternative formulations such as TRADES [16] offer theoretically grounded trade-off mechanisms, this work focuses on standard adversarial training to isolate architectural effects from defense-specific interactions.

E. Training Protocol

a) Clean Training: For CNNs, we use SGD optimizer with initial learning rate 0.1, cosine annealing schedule [53], weight decay 5×10^{-4}, and 200 epochs. For ViTs, we use AdamW optimizer [54] with initial learning rate 10^{-3}, cosine annealing, weight decay 0.05, and 200 epochs.

b) Adversarial Training: Starting from our clean-trained checkpoints, we fine-tune with reduced learning rates: CNNs use SGD with LR 10^{-3} and cosine annealing for up to 100 epochs, while ViTs use AdamW with the same learning rate and schedule. Gradients are clipped to a global L_2 norm of 10, and batches with non-finite loss are skipped. Data augmentation includes random cropping with 4-pixel padding and horizontal flipping; all models consume raw pixel inputs in $[0, 1]$ without mean/std normalization, and attacks and epsilon budgets operate directly in this pixel space. The training batch size is 128 for the CNN and 64 for the ViT (evaluation batch sizes vary by analysis). Early stopping uses patience 20 with a 0.1 percentage-point improvement threshold.

c) Validation Split and Repetitions: Model selection and early stopping use PGD-10 adversarial accuracy on a fixed 2,000-sample validation split. The split is drawn once (seed 777, indices stored with the code) and is shared by every model in the study, and it is excluded from the training set of *both* the clean and the adversarial stage, so no checkpoint is ever selected on data it was trained on. The test set is touched only by the final post-hoc evaluations. We repeat the full pipeline—clean pre-training followed by adversarial fine-tuning—three times per architecture with independent seeds (1001–1003 for

ResNet-18, 2001–2003 for ViT-Tiny), and pair the runs by index so that every cross-architecture comparison is between models trained under the same repetition. All tables report the mean and standard deviation over these three runs.

For one comparison in Section IV-A we also refer to an earlier pair of checkpoints trained before this protocol was adopted, whose validation split had been part of clean pre-training. We report them explicitly as a contrast, not as results, because the difference between the two checkpoint sets is itself evidence about how far a single-run conclusion can move.

F. Evaluation Metrics

a) Robustness Metrics: We evaluate models using three standard metrics: clean accuracy (classification performance on unperturbed test images), robust accuracy (classification performance on adversarial examples), and attack success rate (fraction of correctly classified clean images that become misclassified after attack). For white-box evaluations we additionally report the *conditioned fooling rate*—the fraction of clean-correct samples that the attack flips—and its complement, conditioned survival. Because both PGD and AutoAttack count clean-misclassified samples as non-robust in our per-sample logs, absolute robust accuracy factorizes exactly as robust $=$ clean $\times$ conditioned survival, which lets us decompose robustness differences into a clean-accuracy component and a conditional-susceptibility component. Finally, we report accuracies on the *both-correct* subset (samples that both compared models classify correctly on clean inputs), which yields an exactly paired comparison on a shared sample set.

b) Transfer Attack Metrics: Our primary transfer metric is the *conditioned fooling rate*: among the samples that the target model classifies correctly on clean inputs, the fraction that becomes misclassified under adversarial examples crafted on the source model. Given adversarial examples $\{x^S_{adv}\}$ generated against source model f_S, the conditioned transfer rate to target model f_T is:

$$T_c(f_S \to f_T) = \frac{\sum_{i=1}^{n} \mathbf{1}[f_T(x_i) = y_i]\, \mathbf{1}[f_T(x^S_{adv,i}) \neq y_i]}{\sum_{i=1}^{n} \mathbf{1}[f_T(x_i) = y_i]} \tag{7}$$

Conditioning transfer evaluation on correctly classified samples follows established practice [8], [9], [39]–[41] and is essential in cross-architecture comparisons: the raw misclassification rate $\frac{1}{n}\sum_i \mathbf{1}[f_T(x^S_{adv,i}) \neq y_i]$ includes the target's clean error, so a target with lower clean accuracy appears more "vulnerable" to transfer even when no additional samples are flipped by the attack. We report the raw rate as a secondary metric for transparency and to quantify the size of this confound. Because the conditioning set itself is a protocol choice, we additionally report two established alternatives: the *both-correct* protocol, which conditions on samples that both the source and the target classify correctly [8], [9], and the *successful-source* protocol, which conditions on target-correct samples whose attack succeeded on the source in the white-box sense. As Section IV-B shows, these protocols can yield qualitatively different asymmetry conclusions, so we treat the protocol as part of the reported result rather than an implementation detail. Transfer rates are computed on the full test set (10,000 samples) with per-sample outcomes logged, and we report 95% confidence intervals obtained by percentile bootstrap over samples (10,000 resamples).

c) Gradient Analysis Metrics: We analyze gradient properties through three complementary metric families, computing gradients of per-sample losses so that magnitudes are independent of the evaluation batch size. Gradient norm ($\|\nabla_x \mathcal{L}\|_2$ and $\|\nabla_x \mathcal{L}\|_\infty$) measures overall sensitivity magnitude. Gradient sparsity is quantified with scale-invariant measures—Hoyer sparsity, the Gini coefficient of $|g|$, and the fraction of components below 1% of the per-sample maximum—since fixed absolute thresholds conflate gradient scale with structure; Hoyer and Gini satisfy the scale-invariance axioms formalized by Hurley and Rickard [55]. Gradient alignment measures the average pairwise absolute cosine similarity across samples (computed over all pairs of the 500 analysis samples):

$$\text{Alignment} = \frac{1}{N(N-1)} \sum_{i \neq j} \left| \frac{\nabla_{x_i}\mathcal{L} \cdot \nabla_{x_j}\mathcal{L}}{\|\nabla_{x_i}\mathcal{L}\| \|\nabla_{x_j}\mathcal{L}\|} \right| \tag{8}$$

We use the absolute cosine because shared sensitivity *axes* matter for attack geometry regardless of sign; since sign coherence is what universal-perturbation arguments additionally require, we also report the signed mean cosine, which turns out to be near zero for every model (Section IV-D). High absolute alignment indicates similar perturbation axes across inputs.

d) Spatial Locality Metrics: Sparsity measures how many gradient components are large but says nothing about where they lie in the image, so we measure spatial structure separately on the channel-summed energy map $e = \sum_c (\nabla_x \mathcal{L})_c^2$, normalized to sum to one. We report the smallest pixel fraction whose energy sums to 50% and to 90% (smaller means more localized), the Shannon entropy of e viewed as a distribution over pixels (lower means more localized), and Moran's I with 4-neighbourhood weights, which is positive when energy clusters in adjacent pixels. All four are invariant to the scale of the gradient, and all are computed per sample so that the two architectures can be compared with paired tests on identical inputs.

e) Token Aggregation for Layer-wise Analysis: A transformer block emits one vector per token, so summarizing a block requires a choice. We report three: the CLS token alone, the mean over the 196 patch tokens, and the concatenation of all tokens flattened per sample. For the CNN each residual block output is flattened over its spatial dimensions. We report all three for the ViT because, as Section IV-E shows, they give different drift magnitudes and place the minimum at different depths.

f) Attention Metrics: For the ViT we extract the post-softmax CLS-to-patch attention row of each block, averaged over heads. We report its Shannon entropy, which measures how broadly attention is spread, and its displacement under attack, measured as the total variation distance between the clean and adversarial attention distributions (bounded by 1). Both are computed per sample and averaged, on $n = 1000$ samples per run.

g) ViT-Specific Transfer Attack: To test whether the transfer results depend on using architecture-agnostic attacks, we additionally generate source examples with Token Gradient

Regularization (TGR) [12], which zeroes the extreme-valued entries of the token-gradient tensors in the attention, QKV, and MLP paths during backpropagation and scales the remaining gradient by $\gamma = 0.5$. Our implementation adapts the method to our CIFAR-10 ViT wrapper and to the budget used throughout this paper ($\epsilon = 8/255$, $\alpha = 2/255$, 10 steps, momentum 1.0) rather than the original ImageNet defaults; we therefore report it as an adaptation. Plain MI-FGSM [40] examples are generated in the same run on the same samples, giving a paired control at an identical budget.

h) Feature Degradation Metrics: To analyze how adversarial perturbations affect ViT representations across layers, we compute three metrics at each layer ℓ. Feature cosine similarity measures directional preservation of the representation:

$$\text{Sim}_\ell = \frac{h_\ell(x) \cdot h_\ell(x_{adv})}{\|h_\ell(x)\| \|h_\ell(x_{adv})\|} \tag{9}$$

where h_ℓ denotes the representation at layer ℓ. Feature norm change captures magnitude effects:

$$\Delta_\ell = \frac{\|h_\ell(x_{adv})\| - \|h_\ell(x)\|}{\|h_\ell(x)\|} \times 100\% \tag{10}$$

Feature L_2 distance ($\|h_\ell(x) - h_\ell(x_{adv})\|_2$) quantifies absolute deviation.

G. Statistical Validation

We separate four sources of variance and report each where it applies. *Training-run variance* is the one that bears on architecture comparisons: every table reports the mean and standard deviation over three independent runs per architecture (Section III-E). *Attack-initialization variance* is measured by re-running PGD-10 with three attack seeds (42, 123, 456) on a fixed subset; the evaluated samples and their order are fixed, so the seeds vary only the random start of the attack. *Protocol variance* is the spread across conditioning protocols, which we treat as a result rather than as noise (Section IV-B). *Checkpoint-selection variance* is the spread produced by the offline rule that picks the final checkpoint from a fixed trajectory—validation split, early-stopping patience, and curve smoothing—measured over eighteen combinations while the trajectory itself is held constant. Section IV-G compares their magnitudes.

Paired tests are used wherever the same samples are scored under two conditions: McNemar's exact test for paired binary outcomes, Wilcoxon signed-rank with Holm correction for the three sparsity measures, paired bootstrap confidence intervals (10,000 resamples) for differences, and a sign-flip permutation test (20,000 permutations) for the both-correct transfer difference. Equivalence claims use two one-sided tests (TOST) with the margin stated at each use. All evaluation and analysis scripts fix random seeds (seed 42, including CUDA determinism flags), so every reported number is reproducible from the released pipeline.

IV. Experiments

A. Robustness Comparison

Table I presents the main robustness comparison across all evaluated models. Every cell for our models is the mean over three independent training runs per architecture, with the standard deviation over those runs; each run is evaluated on the full 10,000-sample test set.

TABLE I: Main Robustness Results on CIFAR-10 ($\epsilon = 8/255$), mean±std over three independent training runs per architecture. Each run is evaluated post hoc on the full 10,000-sample test set with fully seeded evaluations (seed 42, deterministic CUDA). Checkpoints were selected on a validation split held out before any training stage (Section III-E). AA = AutoAttack (standard suite).

Model	Defense	Params	Clean	FGSM	PGD-10	AA
CNN Models						
ResNet-18	None	11.2M	95.25±0.15	40.29±1.77	0.03±0.03	–
ResNet-18	AT	11.2M	85.78±0.36	53.46±0.08	44.11±0.50	**37.93±0.14**
WRN-28-10	AT†	36.5M	89.48	70.91	66.92	62.76
Vision Transformer Models						
ViT-Tiny	None	5.7M	80.09±0.60	6.10±1.47	0.05±0.05	–
ViT-Tiny	AT	5.7M	73.53±0.55	36.31±0.42	32.69±0.22	**29.14±0.40**

†Robust checkpoint of Gowal et al. [48] via RobustBench [22]; a single external checkpoint, so no standard deviation applies. Clean/FGSM/PGD-10: our local full-test evaluations; AA: the RobustBench-reported figure (we did not rerun AutoAttack on it).

The CNN holds an *absolute* robustness advantage under every attack: ResNet-18 AT reaches $37.93 \pm 0.14\%$ robust accuracy under AutoAttack against $29.14 \pm 0.40\%$ for ViT-Tiny AT. Because both models of a pair are evaluated on the same 10,000 samples with per-sample outcomes logged, McNemar's paired test applies within each run; it favors the CNN decisively in all three pairs ($p < 10^{-84}$ under AutoAttack, $p < 10^{-120}$ under PGD-10). Seed-level variation is small throughout for the adversarially trained pair, at most 0.55 points on any reported quantity, so the ordering is not a product of run-to-run noise; the undefended baselines vary more under FGSM (up to 1.8 points), but no claim rests on them. The CNN also retains higher clean accuracy (85.78 vs 73.53%, a 12.3-point gap), reflecting the difficulty of training ViTs on small datasets from scratch; Table II separates that contribution from conditional susceptibility. AutoAttack robustness is consistently below PGD-10 robustness for both architectures (6.2 points for ResNet-18, 3.6 for ViT-Tiny), confirming the value of the stronger ensemble. The WideResNet-28-10 reference [48], at 62.76%, shows how far capacity, additional training data, and a stronger recipe jointly push robustness beyond the architectural choice; we keep it visually and analytically separate from the matched pair throughout.

Table II decomposes the absolute scores into their clean-accuracy and conditional-susceptibility components (Section III-F). Because misclassified clean inputs count as non-robust, the factorization is exact: for AutoAttack it reads $85.78 \times 0.442 = 37.93$ for the CNN and $73.53 \times 0.396 = 29.14$ for the ViT. Both factors favor the CNN. It starts from a 12.3-point higher clean accuracy *and* loses a smaller fraction of what it solves, with conditional fooling of 48.58 versus 55.53% under PGD-10 and 55.78 versus 60.37% under AutoAttack. On the jointly solved subset, an exactly paired comparison on shared inputs, the CNN leads by 13.8 and 10.7 points (McNemar $p < 10^{-84}$ in every run).

The decomposition is worth reporting even when the two

TABLE II: Conditional decomposition of white-box robustness on the full test set, mean±std over three runs. Cond. fooling = fraction of clean-correct samples flipped by the attack; absolute robust accuracy factorizes exactly as clean × (1 − cond. fooling). Both-correct columns evaluate the $n = 7{,}061 \pm 68$ samples that *both* models of a pair classify correctly on clean inputs. The lower block reports the same quantities on our earlier single-run checkpoints, whose validation split had been seen during clean pre-training.

Model	Cond. fooling (%) PGD-10	AA	Both-correct robust (%) PGD-10	AA
ResNet-18 AT	**48.58**±0.80	**55.78**±0.23	**59.86**±0.35	**51.89**±0.49
ViT-Tiny AT	55.53±0.50	60.37±0.33	46.11±0.59	41.15±0.39
Earlier single run, validation split seen during clean pre-training (n=7,260)				
ResNet-18 AT	52.15	58.16	**54.92**	**48.15**
ViT-Tiny AT	52.33	**56.46**	49.61	45.33

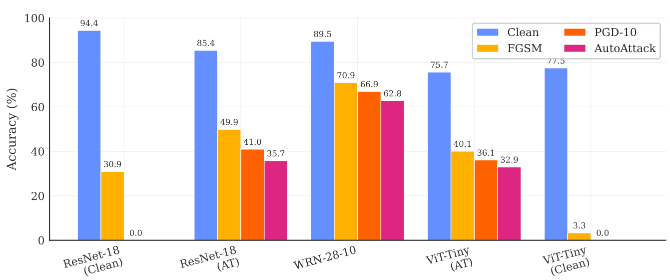


Fig. 1: Clean, FGSM, PGD-10, and AutoAttack accuracy at $\epsilon = 8/255$ on the full CIFAR-10 test set. CNNs outperform ViTs under standard adversarial training; WideResNet-28-10 achieves the highest robust accuracy through capacity, additional training data, and a stronger recipe jointly [48] (its AutoAttack bar is the RobustBench-reported value).

factors agree, because they need not, and because their relative weight is less stable than the ordering they produce. Our first checkpoints, selected with a validation split that clean pre-training had already seen, gave the same absolute ordering but the opposite conditional one: fooling rates that were indistinguishable under PGD-10 (52.15 versus 52.33%) and slightly ViT-favorable under AutoAttack (58.16 versus 56.46%), which supported the reading that the CNN advantage was entirely a clean-accuracy effect (lower block of Table II). None of the three corrected runs reproduces that reading, and the three agree with one another far more closely than any of them agrees with the earlier run. We cannot isolate a single cause, since the corrected protocol changed the validation handling and the training run at the same time; notably, a simple data-quantity explanation does not hold, because the corrected clean models are *better* than the earlier ones despite seeing 2,000 fewer images (80.09 ± 0.60 versus 77.50% for the ViT). The defensible conclusion is narrower and more useful than a mechanism: an attribution of this kind should not be drawn from a single training run, and a paper that reports only the absolute score hides which factor is doing the work.

Figure 1 visualizes this robustness gap across architectures. To further understand the effect of perturbation magnitude, Figure 2 shows PGD-10 robust accuracy across perturbation budgets $\epsilon \in \{2/255, 4/255, 8/255, 16/255\}$ on the full test set.

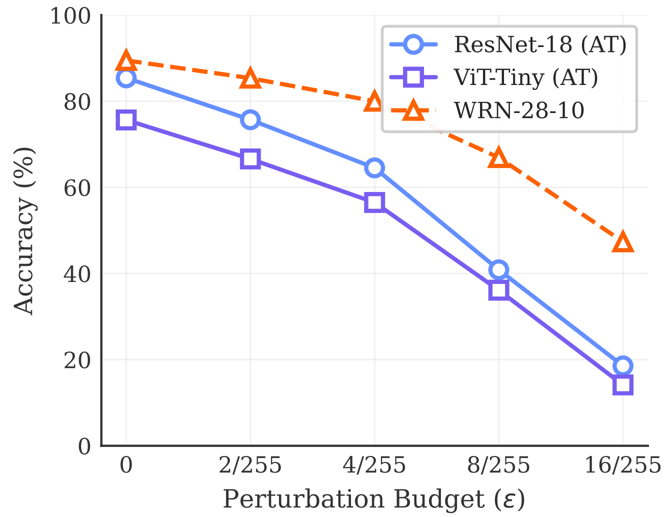


Fig. 2: PGD-10 robust accuracy vs. perturbation budget (ϵ), measured on the full test set. Both AT architectures degrade with similar shapes as epsilon increases, with the CNN maintaining a consistent margin.

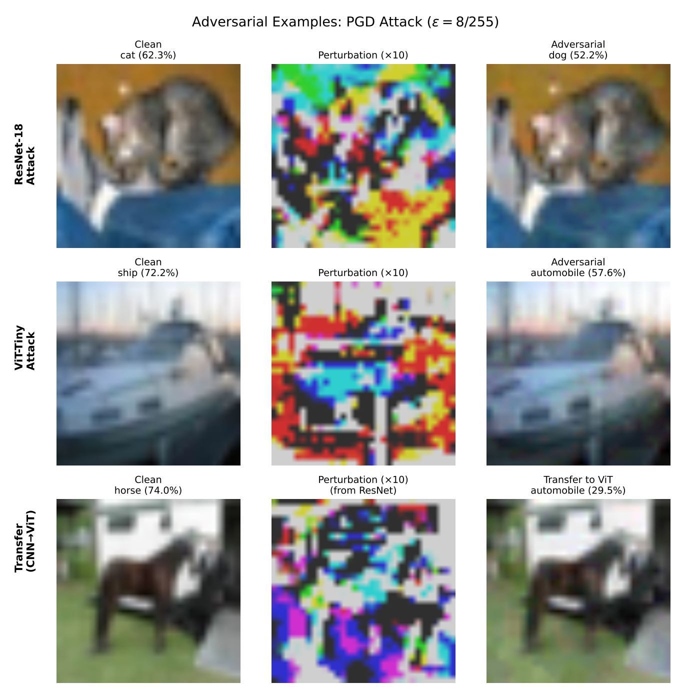


Fig. 3: Adversarial example visualization: for each model, its own white-box PGD-10 attack on the same test image, showing the clean prediction, the perturbation magnified 10×, and the adversarial prediction. Images are displayed with pixel-preserving nearest-neighbour magnification; the perturbations are bounded by ϵ=8/255 and visually subtle at this scale, yet flip the predictions.

Figure 3 provides visual examples of adversarial perturbations and their effects on model predictions.

TABLE III: Transfer asymmetry under the unconditioned rate and three established conditioning protocols (PGD-10, ϵ=8/255, full test set; mean±std over three seed pairs). The rightmost column repeats the difference measured on our earlier single-run checkpoints. The spread on the Diff. column is the standard deviation of the *paired* within-seed difference, which is not recoverable from the two per-direction spreads because the two directions covary across seeds; it is the training-run reference used in Section IV-G. For both-correct the paired estimate computed on the shared subset, quoted in Section IV-B and used for paired inference, is 8.27 ± 0.24; the third-decimal difference from the ± 0.23 shown here reflects only the two estimators. Denominators (CNN→ViT / ViT→CNN): unconditioned 10,000/10,000; target-correct 7,353/8,579; both-correct 7,061 (shared); successful-source 3,122/5,331.

Protocol	**CNN→ViT**	**ViT→CNN**	**Diff.**	**Diff.**[‡]
Unconditioned (raw)	41.02±0.55	27.45±0.27	+13.57±0.33	+8.27
Target-correct	19.87±0.18	15.51±0.59	+4.36±0.44	+0.63
Both-correct	18.25±0.17	9.98±0.31	+8.27±0.23	+5.33
Successful-source	38.50±0.75	23.91±0.80	+14.60±1.48	+5.28

[‡]Single run, validation split seen during clean pre-training.

TABLE IV: 3×3 transfer matrix (PGD-10, ϵ=8/255, full test set; mean±std over three seed pairs; the WideResNet column and row use the single external checkpoint). Raw = unconditioned fooling rate, Cond. = target-correct conditioned rate. Diagonal entries are white-box attacks.

Source	**Target**	**Raw (%)**	**Cond. (%)**
ResNet-18 AT	ResNet-18 AT	55.84±0.53	48.52±0.84
ResNet-18 AT	ViT-Tiny AT	41.07±0.58	19.93±0.20
ResNet-18 AT	WRN-28-10	21.69±0.09	12.48±0.10
ViT-Tiny AT	ResNet-18 AT	27.45±0.34	15.52±0.64
ViT-Tiny AT	ViT-Tiny AT	67.34±0.18	55.57±0.44
ViT-Tiny AT	WRN-28-10	18.25±0.12	8.65±0.14
WRN-28-10	ResNet-18 AT	32.63±0.14	21.48±0.31
WRN-28-10	ViT-Tiny AT	39.73±0.25	18.15±0.28
WRN-28-10	WRN-28-10	33.03	25.16

B. Transfer Attack Analysis

This section contains the paper's central measurement. We craft PGD-10 adversarial examples on a source model and evaluate them on a target, on the full test set, for each of the three seed pairs. Because outcomes are logged per sample, the same attack run can be scored under several conditioning protocols without recomputation: the protocols differ only in which samples enter the denominator (Section III-F).

The result is that the protocol, not the model, dominates the measured quantity. On identical models and identical data, the asymmetry ranges from $+4.36 \pm 0.44$ to $+14.60 \pm 1.48$ points; the mean per-seed spread between the largest and the smallest protocol estimate is 10.45 ± 0.76 points, a factor of 3.3, whereas the seed-level standard deviation within any single protocol stays below 1.5 points. The protocol also decides the statistical verdict. Under the both-correct protocol, which scores both directions on the same samples and therefore supports paired inference, the difference is 8.27 ± 0.24 points with a paired bootstrap CI of $[7.33, 9.21]$, a sign-flip permutation $p < 5 \times 10^{-5}$ in every seed, and TOST equivalence rejected at every margin we tested; under the target-correct protocol on our earlier checkpoints, the same pipeline accepted equivalence at ± 2 points. A reader given only one of these numbers would draw a different conclusion about whether the architectures differ at all.

What is stable is the direction. CNN-crafted examples transfer better to the ViT than the reverse under all four protocols, in all three seeds, and also under the momentum-based MI-FGSM [40] (19.07 ± 0.33 versus $15.42 \pm 0.52\%$ target-correct), so the effect is not an artifact of the PGD attack itself. The magnitude, its significance, and any equivalence claim are protocol-dependent; the sign is not.

The mechanism behind the spread is compositional. Samples that the target solves on clean input but the source does not are three to four times more fragile than jointly solved ones ($41.2 \pm 0.8\%$ and $59.1 \pm 1.7\%$ fooling versus 10.0 and 18.3%), and these marginal subsets differ five-fold in size ($1{,}517 \pm 102$ versus 291 ± 16) because the two models differ in clean accuracy. Conditioning on target-correct admits both marginal subsets, whose asymmetric sizes pull the two rates toward each other; conditioning on both-correct excludes them, and the directional difference reappears.

1) How much of the raw rate is the target's clean error?: To quantify the confound directly rather than argue it, we extend the analysis to a 3×3 matrix by adding the WideResNet-28-10 reference as both source and target (Table IV). Across the six off-diagonal directions, the gap between the unconditioned and the conditioned rate is almost perfectly explained by the clean error of the target: Pearson $r = 0.997$, with a slope of 0.762 points per point of target error. The six directions are three targets crossed with two sources, so the predictor takes only three distinct values and the six points are not independent; an interval computed as if $n = 6$ would overstate the precision. Aggregating to the target level leaves the point estimate essentially unchanged ($r = 0.9985$, same slope) but leaves $n = 3$, where the Fisher-z interval is undefined. We therefore report this correlation as a point estimate over three targets and rest the argument on the slope and its sign rather than on the strength of the correlation. Unconditioned transfer rates therefore do not measure transferability across architectures; they measure transferability plus the target's clean weakness, in a fixed proportion that we can now state.

The third architecture also reframes the asymmetry itself. Averaged over the two foreign sources, incoming conditioned transfer is 18.50% for the ResNet target, 19.04% for the ViT target, and 10.57% for the WideResNet target, which tracks each target's own white-box conditioned fooling rate (48.52, 55.57 and 25.16%) with $r = 0.986$ over the three targets. On this evidence the CNN→ViT advantage is better read as a property of the weaker target than as evidence that CNN-crafted perturbations are intrinsically more transferable: the most robust target absorbs attacks from both families almost equally well. With only three targets this correlation is suggestive rather than conclusive, and we report it as such.

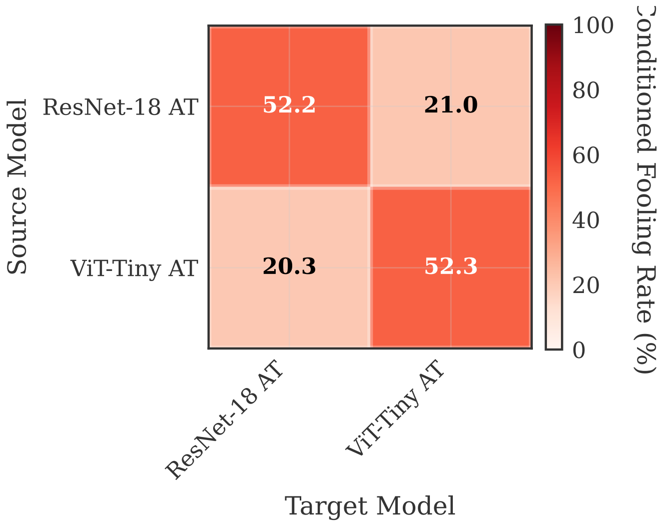


Fig. 4: Conditioned (target-correct) transfer fooling rates, mean over three seed pairs. Off-diagonal asymmetry under this protocol is +4.4 points; Table III shows how far the same quantity moves under other protocols.

TABLE V: CIFAR-100 replication (mean±std over three independent training runs per architecture, full test set, ϵ=8/255). Both models are adversarially trained with the protocol of Section III-E. Absolute robustness is lower than on CIFAR-10 for both architectures, and the CNN-over-ViT ordering is unchanged.

Model	Clean (%)	PGD-10 (%)	AutoAttack (%)
ResNet-18 AT	63.86±1.05	19.30±0.32	15.04±0.54
ViT-Tiny AT	43.17±1.10	11.15±0.60	8.87±0.83

C. Does the Protocol Effect Generalize? A Second Dataset

The measurement above is made on one dataset, and a protocol effect that exists only on CIFAR-10 would not support a general reporting recommendation. We therefore repeated the whole design on CIFAR-100: same architectures, same threat model, same attack budget, same three-seed structure, same four protocols. Before running it we registered three directional predictions and three endpoint bands in an append-only document, so that the outcome could not be reinterpreted after the fact.

The protocol spread grows rather than shrinks. Across the same four protocols the measured CNN→ViT asymmetry runs from $+4.96\pm1.01$ points (target-correct) to $+18.53\pm0.71$ points (unconditioned), with $+10.92\pm0.34$ under both-correct and $+11.44\pm1.82$ under successful-source. The mean per-seed spread is 13.58 ± 1.71 points, against 10.45 ± 0.76 on CIFAR-10. The direction is positive in all twelve measurements (four protocols $\times$ three seed pairs), as it was on CIFAR-10. Under the both-correct protocol the paired difference is 10.92 points with a bootstrap CI of $[9.48, 12.36]$, a sign-flip permutation p below the resolution of 20,000 permutations in every seed, and equivalence rejected at every margin tested.

Two of the three registered predictions are confirmed; the third is not testable in this design. The prediction that the protocol spread would be larger on the harder dataset is confirmed ($13.58 > 10.45$), as is the prediction that the sign would be preserved (12/12). The third predicted that the deviation of the unconditioned from the conditioned rate would grow with the target's clean error. The measured slope is $+0.656$ points per point of target error, positive as predicted; but on CIFAR-100 two of the three targets have clean errors that differ by 0.21 points (36.15 and 36.36), so the correlation is computed over two effective clusters rather than three, and its value ($r = 0.931$ over six directions, 0.974 aggregated to targets) rests on a two-point contrast. The measurement is *consistent with* the prediction and does not *test* it; we report it as such rather than as a third confirmation.

One registered endpoint was missed, and we state why. We predicted clean accuracy in the 33–40% band for the adversarially trained ViT and measured 43.17 ± 1.10%. The band was derived by assuming that the clean-accuracy cost of adversarial training scales the same way for both architectures. It does not: the ViT retained 0.885 of its clean pre-training accuracy under adversarial training, against 0.802 for the ResNet. The two adversarial endpoints, which are the ones the protocol claim depends on, fell inside their bands (PGD-10 predicted 10–14%, measured 11.15; AutoAttack predicted 8–11%, measured 8.87).

Class composition does not manufacture the asymmetry. Because conditioning changes which samples enter the denominator, it also changes the class mix, and an asymmetry could in principle be a composition artifact. We decompose each direction's rate into a composition component and a rate component. Under target-correct the composition effect is -1.249, -1.330 and -1.065 points across the three seed pairs, that is 12.9–18.6% of the asymmetry and *negative*: rebalancing the class mix would make the asymmetry larger, not smaller. Under successful-source it is negligible and its sign is not stable ($+0.039$, $+0.013$, -0.174). Under the unconditioned and both-correct protocols it is exactly zero, which is the arithmetic check the decomposition must pass.

Two qualifications belong with this result. First, the two adversarial bands held with margins of 0.22 points (AutoAttack) and 0.48 points (PGD-10) at the nearest seed-level edge, while the checkpoint-selection ablation reported in Section IV-G measures a selection-protocol amplitude of 1.58 to 2.85 points on a quantity of the same kind. The bands held, but by margins several times smaller than a dispersion we measured ourselves; a reader should weigh the pre-registration accordingly. Second, the validation split used to select the adversarially trained checkpoint is the one already used during clean pre-training. That choice is inherited deliberately from the CIFAR-10 protocol so the two datasets remain comparable, and Section IV-G reports what it does and does not cost.

D. Gradient Characteristics

To characterize architectural differences in input sensitivity, we analyze the gradient properties of both architectures, together with a CIFAR-native ViT control (32×32 inputs, 4×4 patches, no upsampling) trained with the same adversarial protocol, which isolates the effect of the 32→224 upsampling

TABLE VI: Gradient characteristics (n=500 test samples per run; per-sample loss gradients; mean±std over three runs). Alignment is the mean absolute cosine similarity over *all* $500 \times 499/2$ sample pairs. All ResNet–ViT differences on the scale-invariant sparsity metrics are significant under per-sample paired Wilcoxon tests in every run (Holm-corrected $p < 10^{-11}$; paired Cohen's d = 0.32–1.23).

Metric	ResNet-18 AT	ViT-Tiny AT	ViT (native)‡
Hoyer sparsity	**0.4928**±0.0120	0.4561±0.0055	0.405
Gini coefficient	**0.6498**±0.0089	0.6099±0.0051	0.571
Frac. <1% of max	**34.8%**±2.1	26.8%±0.9	17.9%
Alignment (absolute)	0.0378±0.0009	**0.0562**±0.0004	0.079
Alignment (signed)	0.0014	0.0005	–

‡CIFAR-native ViT control (4×4 patches, no upsampling; 5.4M params), a single checkpoint included only to test the upsampling confound.

in our main ViT pipeline. Table VI summarizes the key statistics using scale-invariant sparsity measures (Section III-F); gradients are computed with per-sample losses so that norms do not depend on the evaluation batch size.

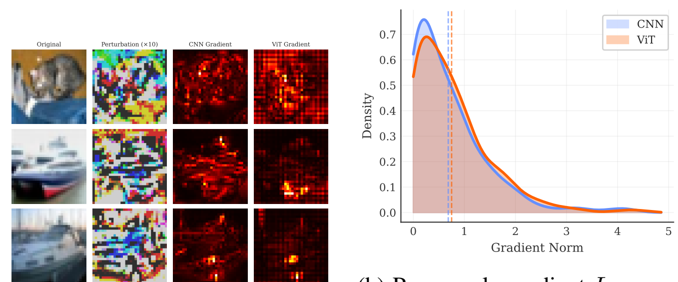
Across all three scale-invariant sparsity measures, CNN gradients are sparser than ViT gradients, and the ordering reproduces in every run with seed-level standard deviations at least three times smaller than the gap. Because the same 500 samples are measured under both models, the differences are tested with per-sample paired statistics: all three are significant in all three runs (Wilcoxon signed-rank, Holm-corrected $p < 10^{-11}$ throughout) with paired Cohen's d between 0.32 and 1.23. The ordering is a product of adversarial training rather than of architecture: in the clean-trained counterparts of the same runs the *ViT* is the sparser model (Hoyer 0.380 ± 0.009 versus 0.347 ± 0.002), so adversarial training sparsifies CNN gradients far more strongly, consistent with [27]. The native-resolution ViT control is less sparse than the upsampled ViT on every measure, so the ordering is not an artifact of the $32 \to 224$ upsampling. Cross-sample alignment is higher for the ViT in absolute cosine, and the clean-trained pair already shows this gap (0.058 versus 0.029), which makes it a property of the architecture rather than of the defense. The *signed* mean cosine is near zero for both models (≤ 0.0014): what samples share is sensitivity axes, not consistent directions, so the alignment gap is not evidence for cheap universal perturbations.

These gradient results are also the one part of our analysis that is insensitive to which checkpoint set is used: every quantity in Table VI reproduces the value measured on our earlier single-run checkpoints to within the reported standard deviations, in contrast to the conditional decomposition of Section IV-A.

1) Sparsity is not spatial locality: Sparsity measures answer how *many* gradient components are large, not whether those components sit together in the image. The distinction matters because the sparser-CNN result is frequently described in spatial language—gradients said to be more "localized" or "concentrated" on object regions—and that reading implies a claim these measures cannot support. We therefore measure spatial structure directly on the same gradients, using four scale-invariant descriptors of the channel-summed energy map:

TABLE VII: Spatial locality of input gradients (n=500 samples per run; mean±std over three runs). Paired differences are computed on the same samples; p is the largest per-run paired Wilcoxon p-value. Lower area and entropy mean more localized; higher Moran's I means more clustered.

Measure	ResNet-18	ViT-Tiny	Diff.	p
Area for 50% energy	0.0378±0.0017	0.0397±0.0010	−0.0019	0.26
Area for 90% energy	0.2345±0.0078	0.2570±0.0067	−0.0224	0.039
Spatial entropy (nat)	5.2014±0.0387	5.2524±0.0197	−0.0509	0.80
Moran's I	0.4141±0.0132	0.3941±0.0024	+0.0200	0.54



(a) Gradient field visualization

(b) Per-sample gradient L_2 norm distribution

Fig. 5: Gradient characteristics comparison. (a) Input-gradient magnitude maps for representative test images with the corresponding PGD-10 perturbations. (b) Distribution of per-sample gradient L_2 norms for both architectures.

the smallest pixel fraction carrying 50% and 90% of the gradient energy, the spatial entropy of the normalized energy map, and Moran's I with 4-neighbourhood weights, which is positive when energy clusters in adjacent pixels.

The result is negative (Table VII). Three of the four measures show no significant difference, and the one that reaches nominal significance, the area carrying 90% of the energy, points in the direction of a *smaller* area for the CNN by two percentage points of image area, which is a marginal effect next to a Hoyer gap that is significant at $p < 10^{-11}$. Neighbourhood clustering is essentially identical for the two architectures. The two families of measures therefore disagree, and the disagreement is informative: adversarially trained CNN gradients place more of their mass on fewer pixels, but those pixels are not more spatially grouped than the ViT's. Descriptions of CNN gradients as spatially localized are not supported by our measurements, and we restrict our own claims to sparsity.

Figure 5 provides visual evidence for these gradient differences.

Figure 6 shows a direct comparison of gradient heatmaps, highlighting the structural differences in how each architecture computes input gradients.

E. Feature Degradation in Vision Transformers

We next measure where adversarial damage accumulates along depth, extending layer-wise representation analyses of clean-trained ViTs [28] to the adversarially trained setting. Two design choices in this measurement are usually left implicit and turn out to matter, so we vary both: the sample

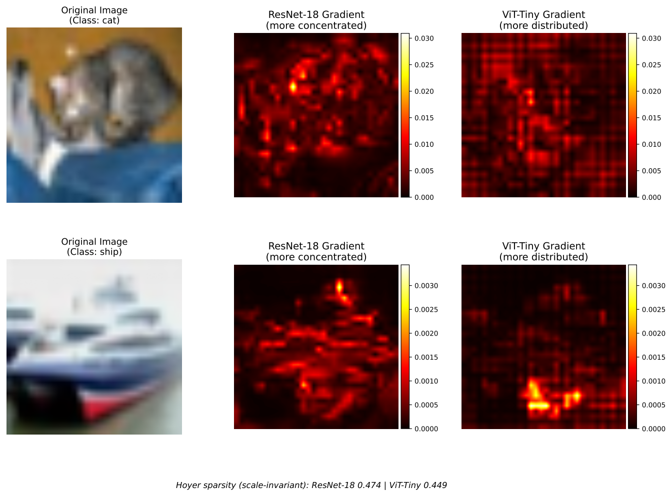


Fig. 6: Gradient heatmap comparison between CNN and ViT on the same input images. CNN gradients (top) are modestly more concentrated around object edges, while ViT gradients (bottom) are somewhat more spatially distributed (Hoyer sparsity annotations from the gradient-analysis artifacts).

TABLE VIII: Block-wise clean–adversarial cosine similarity under each model's own white-box PGD-10 attack (n=1000 per run; mean$\pm$std over three runs). ViT columns differ only in how a block's tokens are aggregated.

	ViT-Tiny AT			**ResNet-18 AT**	
Blk	**CLS**	**Patch mean**	**All tokens**	**Block**	**Cosine**
1	0.9964	0.9990	0.9975	layer1.0	0.9951
2	0.9893	0.9954	0.9911	layer1.1	0.9918
3	0.9785	0.9928	0.9821	layer2.0	0.9879
4	0.9709	0.9877	0.9734	layer2.1	0.9853
5	0.9652	0.9870	0.9719	layer3.0	0.9753
6	0.9596	0.9873	0.9712	layer3.1	0.9607
7	0.9497	0.9870	0.9698	layer4.0	**0.8777**
8	0.9414	**0.9840**	**0.9663**	layer4.1	0.9108
9	0.9353	0.9850	0.9675		
10	**0.9343**	0.9877	0.9696		
11	0.9352	0.9912	0.9753		
12	0.9779	0.9955	0.9884		

size ($n = 1000$ per run rather than a single small batch) and the token aggregation used to summarize a transformer block. For the ViT we report the CLS token alone, the mean over the 196 patch tokens, and the flattened concatenation of all tokens; for the CNN each residual block output is flattened over its spatial dimensions.

Two results follow. First, the two architectures differ less in *whether* damage accumulates than in *where*. In the ViT the drift is gradual and shallow: similarity declines steadily to a minimum around blocks 8–10 and then recovers toward the output, while feature norms stay within 0.7% of their clean values throughout, so the perturbation rotates the representation rather than rescaling it. In the CNN the damage is concentrated: the first three stages lose little, then the first block of the last residual stage collapses to 0.8777 ± 0.0181 together with a $-13.14 \pm 1.89\%$ norm contraction, after which the final block partially repairs the representation (0.9108 ± 0.0060, norm $+5.84 \pm 3.53\%$). Adversarial damage in the CNN is thus a late, localized, magnitude-changing event; in the ViT it is a distributed, direction-only rotation. We note that on our earlier checkpoints the CNN profile fell monotonically to the last block, with no recovery, so the shape of the CNN curve—unlike the ViT profile—should be read as recipe-dependent.

Second, the aggregation choice changes the measured profile substantially. The CLS pathway drifts about three times as far as the patch-token mean (minimum 0.9343 versus 0.9840), and the two summaries even place the minimum at different depths (block 10 versus block 8). Since the CLS token is what the classifier reads, this is not a nuisance but a result: the perturbation concentrates on the classification pathway while the average patch representation is comparatively preserved. It also means that layer-wise drift curves from different papers are not directly comparable unless the aggregation is stated.

A third measurement is negative. Adversarial perturbation does not measurably change the entropy of the CLS attention distribution: over $n = 1000$ samples per run, the mean per-layer change is at most 0.005 nats in absolute value against layer means near 4.9 nats, with a seed-level standard deviation of the same order. What does grow with depth is the *displacement* of the attention map, from 0.028 total variation in the first layer to 0.084 in the last—small in absolute terms, since total variation is bounded by 1. The attack therefore redirects attention slightly without broadening or sharpening it. We flag this explicitly because an earlier version of this analysis, computed on an 8-sample figure batch, suggested a visible attention effect that the larger measurement does not support; claims of "attention degradation" under adversarial attack should be checked at this sample size before being made.

Finally, drift magnitude does not predict attack success. Splitting the samples by whether the attack flipped the prediction, the last-block cosine similarity of flipped and unflipped ViT samples is statistically indistinguishable (0.9895 versus 0.9877); for the CNN there is a small difference in the expected direction (0.9058 versus 0.9143). Feature drift is therefore a description of how the representation moves, not a sufficient statistic for whether the prediction changes.

Complementing the quantitative measurements, Figure 7 visualizes CLS-token attention maps at layers 1, 6, and 12 for a representative sample. The visualization is consistent with the displacement measurement above—later layers move more than early ones—but we caution that single-sample attention panels of this kind are easy to over-read: the aggregate entropy change over 1,000 samples is null, so the visible differences reflect redirection, not diffusion. Attention-shift effects under patch perturbations [31] and attacks that explicitly target the attention mechanism [30] operate in a different regime from the L_∞-bounded perturbations studied here.

Figure 8 shows the clean and adversarial CLS attention for a representative sample together with the per-layer entropy curves; the two entropy curves lie on top of each other, which is the visual form of the null result reported above.

Figure 9 shows an illustrative t-SNE embedding of the penultimate feature spaces. Because t-SNE embeddings are sensitive to hyperparameters and invite qualitative over-reading, we base all claims on metrics computed in the original

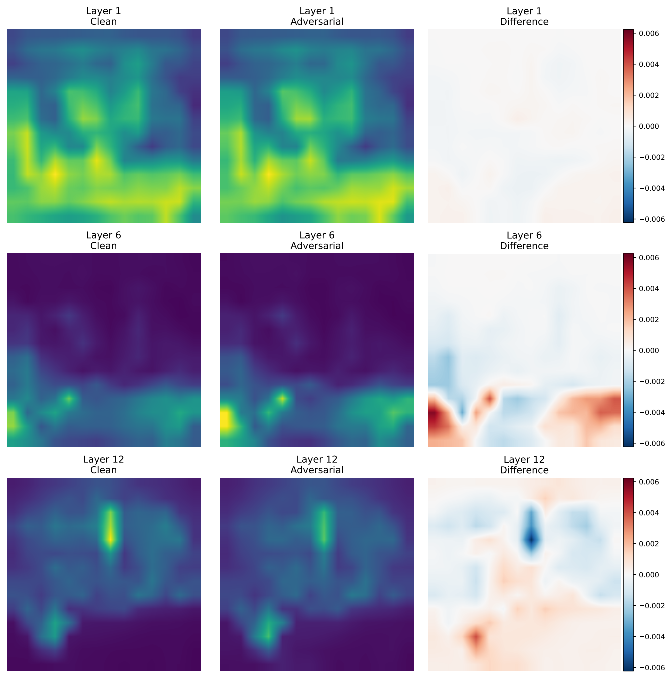
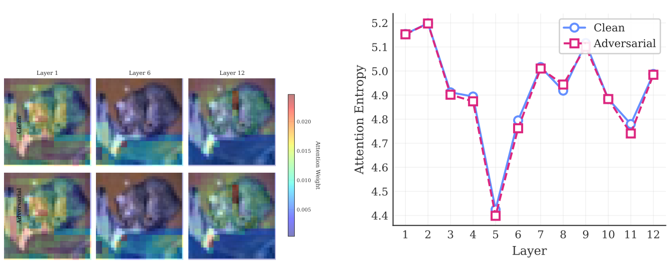


Fig. 7: CLS-token attention maps (post-softmax, head-averaged) across ViT layers under adversarial attack. Each row shows a different layer (1, 6, 12); columns show clean attention, adversarial attention, and their difference. Early layers maintain similar attention patterns, while later layers show larger attention shifts.



(a) Attention comparison (b) Layer-wise attention entropy

Fig. 8: ViT attention analysis under attack: (a) clean vs. adversarial CLS attention for a representative sample; (b) per-layer entropy of the CLS attention distribution for clean and adversarial inputs.

feature space (n=500 per run, each model attacked white-box, mean±std over three runs). The attack collapses class structure most strongly in the *CNN*: clean-fit 5-NN label consistency drops from $87.3 \pm 2.6\%$ to $51.3 \pm 1.8\%$, the silhouette score falls from 0.196 ± 0.010 to -0.032 ± 0.004, the intra/inter-class distance ratio rises from 0.511 ± 0.013 to 0.847 ± 0.003, and the relative class-centroid shift is 0.238 ± 0.012. The ViT's penultimate space is only weakly class-separated even on clean inputs (silhouette -0.021 ± 0.005; 5-NN consistency $61.1 \pm 1.4\%$, dropping to $48.5 \pm 1.4\%$), and its centroids shift far less (0.083 ± 0.004). The measurable statement is therefore that the CNN's well-separated clean feature space is heavily disrupted by white-box attack, whereas the ViT's

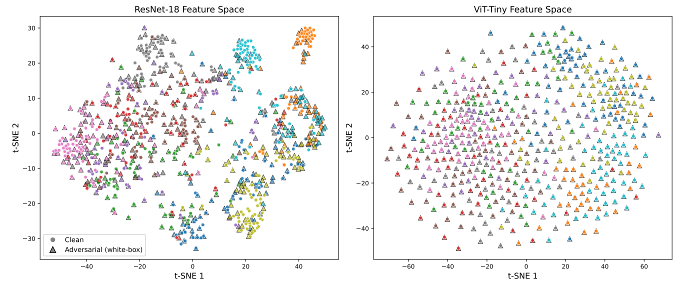


Fig. 9: Illustrative t-SNE embedding of penultimate features for clean samples (circles) and white-box PGD-10 adversarial samples (triangles), for ResNet-18 (left) and ViT-Tiny (right). t-SNE is shown for orientation only; the quantitative comparison is given in the text.

TABLE IX: ViT→CNN transfer under TGR [12] and plain MI-FGSM at a matched budget (n=10,000 per run; mean±std over three runs).

Measure (%)	**TGR**	**MI-FGSM**
White-box on source (raw)	51.56±0.80	**66.43**±0.29
Transfer, target-correct	12.70±0.41	**15.71**±0.60
Transfer, both-correct	7.93±0.42	**10.08**±0.30

class structure is weaker to begin with and moves less in these aggregate terms; this is a statement about geometry, not about robustness, since the CNN nonetheless retains the higher robust accuracy.

F. *Does a ViT-Specific Transfer Attack Change the Picture?*

All transfer results so far use gradient attacks that treat the ViT as a generic differentiable model. A natural objection is that the ViT direction is handicapped by this choice, since attacks designed for transformer structure are reported to transfer better. We test the objection directly with Token Gradient Regularization (TGR) [12], which suppresses extreme-valued token gradients in the attention, QKV, and MLP paths during backpropagation on the source model. Our implementation is an adaptation: the original targets standardly trained ImageNet ViTs, whereas we apply the same mechanism to our CIFAR-10 ViT-Tiny wrapper and match the attack budget to the rest of the paper ($\epsilon = 8/255$, $\alpha = 2/255$, 10 steps, momentum 1.0) rather than using ImageNet defaults. As a control we generate plain MI-FGSM examples in the same run, so the comparison is paired on identical samples and identical budget.

TGR does not improve transfer in this regime; it degrades it (Table IX). The regularization behaves as designed on the source side, trading away 14.9 points of white-box strength, but the trade does not pay off at the target: conditioned transfer falls by 3.0 points under the target-correct protocol and by 2.2 under both-correct. Paired McNemar tests on the both-correct subset favor MI-FGSM decisively in every run ($p = 6.1 \times 10^{-10}$, 1.1×10^{-27}, 7.0×10^{-34}), so this is not a seed effect.

We state the conclusion narrowly. Our setting differs from the one TGR was designed for in three respects that could each matter: the targets are adversarially trained rather than standardly trained, the data are 32×32 CIFAR-10 images upsampled inside the model rather than native ImageNet, and the source is a 5.7M-parameter ViT-Tiny. The claim we can support is that TGR's transfer advantage does not carry over to this regime, not that the method fails in general. For the purposes of this paper the relevant consequence is that the CNN$\rightarrow$ViT asymmetry reported in Section IV-B is not an artifact of using architecture-agnostic attacks: giving the ViT direction a transformer-specific attack makes it weaker, not stronger.

G. Sources of Variance

Four sources of variance bear on the comparisons above. Attack-initialization stochasticity is negligible: re-running PGD-10 with three attack seeds (42, 123, 456) on a fixed 1,000-sample subset leaves clean accuracy exactly unchanged, since the loader does not shuffle, and moves robust accuracy by a standard deviation of 0.05 points for the CNN and 0.00 for the ViT. Training-run variance is larger but still small: across the three independent runs per architecture, the standard deviation for the adversarially trained pair is at most 0.55 points on clean accuracy and 0.50 on PGD-10 (Table I).

Comparing the protocol effect against those two figures directly would be an error of scale, because the protocol effect is measured on the transfer asymmetry while both figures above are measured on absolute accuracy. We therefore quantify the protocol and training-run effects on the *same* quantity, the off-diagonal asymmetry, and report both in absolute units. Within a fixed seed pair, changing only the conditioning protocol moves the asymmetry by 10.45 ± 0.76 points. This figure is the mean of the per-seed ranges and slightly exceeds the 10.24-point range of the protocol means, because the extremal protocol is not the same in every seed: in the third seed pair the largest asymmetry comes from the unconditioned rate rather than from successful-source. Retraining both architectures from scratch moves that same asymmetry by a standard deviation of only 0.23 to 1.48 points, the value depending on which protocol it is measured under (Table III, Diff. column). A ten-point measurement effect against a sub-1.5-point training effect is the ordering the paper rests on.

We deliberately do not headline the ratio of these two spreads. With three runs the denominator carries two degrees of freedom, so a χ^2 interval on any single ratio is very wide: for the successful-source protocol, whose run-to-run standard deviation is the largest of the four, the 95% interval on the ratio runs from 0.5 to 6.3 and therefore includes unity. Reported as a sensitivity rather than an estimate, the ratio spans 3.3 to 22.7 across dispersion measures and reference protocols, and equals 20.9 under the like-for-like standard-deviation comparison on the both-correct protocol that we treat as primary. Two cautions bound it further. The protocol with the largest asymmetry also carries the largest run-to-run standard deviation, so numerator and denominator are not independent. And because all three runs share the single validation split drawn before any training, this standard deviation excludes split-to-split variance, which makes the ratio an upper bound. The absolute comparison survives all three caveats, which is why it, and not the fold-change, carries the claim. This ordering is the quantitative form of the paper's argument: with a protocol left unstated, a reader cannot tell a real architectural difference from a measurement choice.

A fourth source is the offline checkpoint-selection protocol, which is usually left implicit. Holding a training trajectory completely fixed and varying only how its final checkpoint is chosen—which validation split, what early-stopping patience, and whether the validation curve is smoothed before the argmax (eighteen combinations)—moves the reported PGD-10 accuracy by 2.62 to 2.85 points per seed for the ResNet and 1.58 to 2.09 points for the ViT. The weights on disk are identical across these cells; only the rule that picks one of them differs. We report this dispersion in absolute units and deliberately not as a ratio to seed-level dispersion, because that ratio depends on which of the eighteen cells is taken as the reference: swept across all eighteen it runs from 0.67 to 3.38 for the ResNet and 0.63 to 7.38 for the ViT, falling below unity for some references and inverting the comparison.

A counterweight belongs immediately next to it. On CIFAR-100 the epoch selected by our own protocol varies by 25 epochs across the three ResNet seeds, yet the resulting test PGD-10 standard deviation is 0.32 points. The selection *path* was unstable while the selected *outcome* was not. Neither observation generalizes without the other: selection choices can move a reported number by more than a training seed does, and they can also leave it nearly untouched, and which of the two happens is a property of the trajectory's plateau rather than of the architecture. The practical consequence for reporting is the same in both cases—the selection rule is part of the result and belongs in the methods section.

Computational Details: The training runs and analyses reported here were conducted on an NVIDIA RTX 5090 (32GB VRAM); the earlier single-run checkpoints used for the comparison in Table II were trained on an RTX 5060 Ti (16GB). All runs used PyTorch 2.6.0 with CUDA 12.8 and deterministic cuDNN settings. AutoAttack evaluation used the standard configuration (APGD-CE, APGD-T, FAB-T, Square) on the full 10,000-sample test set in seeded 1,000-sample chunks with per-sample outcome logging. The three-seed training pipeline required 21.7 GPU-hours in total.

V. Discussion

A. What the Measurements Show

a) Protocol variance dominates the quantities being compared: The central result is an ordering of dispersions. Measured on the transfer asymmetry itself, retraining both architectures from scratch moves it by a standard deviation of 0.23 to 1.48 points, whereas changing only the conditioning protocol moves it by 10.45 ± 0.76 points (Section IV-G). We state the two spreads in absolute units rather than as a fold-change, because with three runs the interval on that ratio is too wide to carry a headline. For scale on a different quantity, re-running an attack with a new random initialization moves absolute robust accuracy by about 0.05 points and retraining moves the adversarially trained pair by at most 0.55.

A quantity that the measurement choice moves by ten points while retraining moves it by one is not a stable property of the architectures, and reporting it without the protocol communicates almost nothing. This is not an argument against conditioning—conditioning on correctly classified samples is established practice [9], [39], [40] and remains necessary—but an argument that the choice belongs in the result, not in a footnote.

The four protocols are not interchangeable, and each has a defensible use. The unconditioned rate should not be used for architecture comparison at all: our 3×3 matrix shows its deviation from the conditioned rate is explained by the target's clean error with $r = 0.997$ over three targets, which means it measures target weakness in fixed proportion rather than transferability. The target-correct protocol removes that confound but admits samples the source itself cannot solve, and because those marginal subsets are hyper-vulnerable and asymmetrically sized, it pulls two directions toward each other by an amount that depends on the clean-accuracy gap. The both-correct protocol scores both directions on identical samples and is the only one of the four that supports paired inference, which is why we treat it as the primary estimate. The successful-source protocol answers a different question—how well an attack that already worked carries over—and its larger value reflects that conditioning on source success selects strong perturbations.

b) The direction is stable; the magnitude and the verdict are not: Across four protocols, three training runs, and two attack families (PGD-10 and MI-FGSM), adversarial examples crafted on the CNN transfer better to the ViT than the reverse. What changes is how large the effect looks and whether an equivalence test accepts it. Our three-target analysis suggests a reading of this direction that does not require the CNN's perturbations to be special: incoming transfer tracks each target's own white-box vulnerability ($r = 0.986$ over three targets), and the most robust target absorbs attacks from both families almost equally. On this account the asymmetry is a statement about which model is the weaker target, which is consistent with our finding that a transformer-specific attack [12] does not rescue the ViT direction: with three targets the correlation is suggestive, and a larger model pool would be needed to establish it.

c) Attribution within a headline number is less stable than the headline: The conditional decomposition separates absolute robustness into a clean-accuracy factor and a conditional-susceptibility factor. In our corrected runs both favor the CNN, so the decomposition and the headline agree. They did not agree on our earlier checkpoints, where conditional susceptibility appeared matched under PGD-10 and slightly ViT-favorable under AutoAttack. Since the absolute ordering was identical in both checkpoint sets, a paper reporting only absolute accuracy would have seen nothing change, while a paper reporting the decomposition from a single run would have drawn a mechanistic conclusion that does not replicate. The practical lesson is that mechanism-level claims need more training runs than ranking-level claims, and the two should not be given the same evidential weight.

d) Three claims that did not survive measurement: Sparsity is not spatial locality. Adversarially trained CNN gradients concentrate more mass on fewer components (Hoyer 0.4928 versus 0.4561, paired $p < 10^{-11}$), but four spatial descriptors of the same gradients show no corresponding difference in how those components are arranged in the image. Descriptions of CNN gradients as spatially localized therefore need separate evidence; sparsity measures [55] do not provide it.

Adversarial perturbation does not measurably change CLS attention entropy. At $n = 1000$ the per-layer entropy change is at most 0.005 nats, while attention maps are displaced by a small and depth-increasing amount. Attention-shift results obtained with patch attacks [30], [31] concern a different threat model and should not be generalized to L_∞-bounded perturbations without measurement.

A ViT-specific transfer attack does not improve ViT-sourced transfer here. TGR [12] costs 14.9 points of white-box strength and returns none of it at the target in our regime. We report this as a regime-specific negative result, since our targets are adversarially trained and our data are CIFAR-scale, and it is exactly the kind of claim that a single untested assumption would otherwise import into a comparison.

e) What survives: Two behavioral differences reproduce across independent checkpoint sets and are unaffected by the protocol question. Gradient structure differs: CNN input gradients are sparser under all three scale-invariant measures, adversarial training rather than architecture creates that ordering (the clean-trained pair has it reversed), and cross-sample alignment is higher for the ViT in absolute cosine while the signed mean is near zero for both—samples share sensitivity axes, not directions, which is the distinction that separates our measurement from a universal-perturbation claim [56]. Depth profiles differ: adversarial damage in the CNN is a late, localized, magnitude-changing event confined to the last residual stage, whereas in the ViT it is a distributed, direction-only rotation. That the ViT's measured profile depends on token aggregation—the CLS pathway drifts three times as far as the patch-token mean—is itself a reporting requirement for layer-wise studies.

B. Reporting Requirements

The measurements above imply a short and concrete list for cross-architecture robustness studies.

1) **State the conditioning protocol, and prefer paired ones.** Report which samples enter the denominator. Where a directional claim is made, use a protocol that scores both directions on identical samples, so paired inference is available.
2) **Do not report unconditioned transfer rates as architecture comparisons.** When targets differ in clean accuracy, the raw rate is a fixed mixture of transferability and target weakness; our estimate of the mixing is $r = 0.997$, from three targets.
3) **Report the decomposition, not only the score.** Absolute robustness factors exactly into clean accuracy and conditional susceptibility; which factor carries a headline

is the interesting part and is not recoverable from the score.

4) **Separate ranking claims from mechanism claims.** Our rankings were stable across checkpoint sets; our mechanism attribution was not. Mechanism claims should carry more training runs.
5) **State the aggregation for layer-wise analyses.** CLS-only, patch-mean, and all-token summaries of the same blocks give different drift magnitudes and different minima.
6) **Check attention and locality claims at a real sample size.** Both of the null results we report would have looked positive on a small figure batch.

For practitioners choosing between the two families under matched standard adversarial training on this benchmark, the CNN gives higher absolute robust accuracy and the advantage holds under every view we measured. The WideResNet reference at 62.76% shows that recipe, capacity, and extra data jointly [48] move robustness further than the architectural choice does, which is worth weighing before investing in architecture selection as a robustness strategy.

C. Limitations

Our study is deliberately narrow, and the narrowness bounds the conclusions in specific ways.

The protocol-sensitivity result is measured on two datasets and one model pair. On CIFAR-100 the spread is larger than on CIFAR-10 (13.58 ± 1.71 against 10.45 ± 0.76 points) and the sign is preserved in all twelve measurements (Section IV-C), which is what the mechanism predicts, since it follows from a clean-accuracy difference between targets rather than from anything CIFAR-specific. What we still cannot state is the functional form of the dependence: two datasets give two points, and on CIFAR-100 two of the three targets have nearly identical clean error, so the within-dataset slope rests on a two-point contrast. A study spanning several architecture pairs with varying clean-accuracy gaps would turn these measurements into a calibration curve.

We cannot isolate the cause of the differences between our two checkpoint sets. The corrected protocol changed the validation handling and produced a fresh training run at the same time. A data-quantity explanation is ruled out, since the corrected clean models are better despite less data, but distinguishing selection leakage from ordinary run-to-run variation would require an ablation that trains on identical data with and without a leaked selection split. We report the contrast as evidence that single-run mechanism claims are fragile, not as a measurement of leakage.

Every measurement in this paper is made inside a single threat model: L_∞ at $\epsilon = 8/255$. Whether the protocol spread survives a change of norm is a separate question, and one we have registered but not yet answered. It is worth stating what such a test could and could not show: our models are L_∞-trained, so evaluating them under an L_2 budget probes the measurement axis rather than L_2 robustness, and the resulting absolute values would not be comparable with L_2-trained references.

Three runs per architecture bound seed variance but do not span the recipe space, and ViT adversarial training is known to be recipe-sensitive on small datasets [50], [51]; with diffusion-generated synthetic data ViTs have been reported to surpass WideResNets on CIFAR-10 [35], [57]. Our ViT results characterize baseline adversarial training, not the best achievable ViT robustness. The ViT-Tiny pipeline upsamples 32×32 inputs to 224×224 inside the model; our CIFAR-native control shows the gradient ordering is not an upsampling artifact, but that control differs in capacity and patch size and therefore calibrates the confound rather than providing a second matched pair. The two models also differ in parameter count (11.2M versus 5.7M), so architectural attribution of the absolute gap remains partial. Finally, we evaluate a single-budget L_∞ threat model, and our TGR result is an adaptation of the original method rather than a reproduction of its published setting.

D. Future Directions

The most direct extension is to turn the protocol-sensitivity measurement into a general one: sweep architecture pairs with controlled clean-accuracy gaps and measure how the spread across protocols scales with that gap. If the relationship is as regular as our three-target correlation suggests, it would yield a correction that lets published raw-rate asymmetries be reinterpreted rather than discarded.

Two of the extensions we previously listed as future work have since been run, and we record both what they settled and what they did not. The leakage ablation was carried out on six full-budget trajectories, and it did not detect an effect of leakage on checkpoint selection; it answers a weaker question than the one originally posed, because the treatment was applied to a single validation split, so split identity and leakage cannot be separated within that design. Its informative by-product is the selection-protocol dispersion reported in Section IV-G. The second dataset has also been run (Section IV-C). What remains open is the capacity-matched pair (for example ViT-Small against ResNet-50), which would test whether the surviving behavioral differences—the sparsity ordering and the depth profiles—are properties of the families or of these particular small models.

VI. Conclusion

We asked how much of the disagreement in CNN–Vision Transformer robustness comparisons comes from the measurement rather than from the models, and answered it by holding the models, the data, the threat model, and the attack budget fixed while varying only how the comparison is scored.

The measurement dominates. On identical adversarially trained ResNet-18 and ViT-Tiny pairs, the cross-architecture transfer asymmetry ranges from $+4.4$ to $+14.6$ points across four established conditioning protocols, a 3.3-fold spread, whereas retraining both architectures from scratch moves that same quantity by under 1.5 points, and the protocol decides whether an equivalence test is accepted or rejected. Unconditioned rates are not a valid basis for such a comparison at all: in a 3×3 matrix that adds a WideResNet-28-10 reference, their deviation from conditioned rates is explained by the target's clean error with $r = 0.997$ over three targets. The same design run on CIFAR-100 widens the spread to 13.58 ± 1.71 points rather than shrinking it, and an ablation that fixes the trajectory and varies only the checkpoint-selection rule moves reported robust accuracy by 1.58 to 2.85 points, so measurement choices that are usually left implicit are of the same order as the effects being reported. What the protocol does not change is the direction, which is stable across protocols, seeds, datasets, and attack families, and which our three-target analysis associates with the target's own vulnerability ($r = 0.986$) rather than with any special potency of CNN-crafted perturbations.

A conditional decomposition separates absolute robustness into a clean-accuracy factor and a conditional-susceptibility factor. Both favor the CNN in our corrected runs, but the attribution between them differed on an earlier checkpoint set whose absolute ordering was identical, which shows that mechanism-level claims are more fragile than ranking-level claims and need more training runs to support.

Three claims common in this literature did not survive direct measurement: adversarially trained CNN gradients are sparser but not spatially more localized, CLS attention entropy does not change under L_∞ attack at $n = 1000$, and a transformer-specific transfer attack yields no transfer gain in this regime. Two behavioral differences did survive and reproduce across independent checkpoint sets: the gradient sparsity ordering, which adversarial training creates rather than architecture, and the depth profile of adversarial damage, which is a late localized magnitude-changing event in the CNN and a distributed direction-only rotation in the ViT.

From these measurements we distilled six reporting requirements, of which the first two matter most: state the conditioning protocol and prefer paired ones, and do not present unconditioned transfer rates as architecture comparisons. Our study covers one dataset and one model pair, so the size of the protocol effect in other settings remains to be calibrated; the mechanism behind it, an unequal clean-accuracy gap between targets, is not specific to our setting. We release the full pipeline, per-sample logs, and analysis scripts so that the protocol comparison can be reproduced and extended.

Data and Code Availability

The experiments in this study use the CIFAR-10 dataset [11], which is publicly available at https://www.cs.toronto.edu/~kriz/cifar.html. Pre-trained WideResNet-28-10 weights were obtained from the RobustBench benchmark [22], [48]. The source code for the analysis pipeline and trained model checkpoints will be made publicly available upon acceptance.

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